Advanced Pattern Matching Strategies for SP500 Bump & Slide Analysis

1. The Limitation of Current "Magnitude-Only" Matching

The current "Goal Seek" engine identifies patterns by measuring the net price change over a fixed window: `Price(T+N) - Price(T)`.

The Problem: This ignores the "journey" the price took within that window.

- **False Negatives:** A strong move that hits a peak and retraces slightly by the end of the window might be discarded because the net change is below the threshold.
- **False Positives:** A highly volatile, "choppy" move that happens to end higher but shows no consistent trend would be included.
- **Boundary Sensitivity:** If the window is 15 minutes, but the actual "Bump" lasted 18 minutes, the pattern is cut off or diluted.

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2. Proposed Intelligent Pattern Strategies

To move beyond simple magnitude change, we can implement vectorized "Shape and Quality" metrics. These remain high-performance and compatible with the existing search engine.

Strategy A: Trend Efficiency (Kaufman's Efficiency Ratio)

Instead of just net change, we measure the "straightness" of the move.

- **Concept:** `Net Change / Sum of Absolute One-Minute Changes`.
- **Logic:** A ratio of 1.0 means the price went straight up every minute. A ratio of 0.2 means there was massive "noise" and retracement during the move.
- **Benefit:** Filter for "Clean Bumps" (e.g., `Efficiency > 0.6`) to ignore choppy, random price action.

Strategy B: Maximum Excursion (Peak/Trough Recognition)

Capture the "true" peak of the move, regardless of where the fixed window ends.

- **Concept:** Measure the `Maximum Favorable Excursion (MFE)`—the highest price reached *anywhere* inside the bump window.
- **Logic:** A Bump is valid if the *Peak* reached a certain threshold, even if the price settled lower by the end of the 15-minute window.
- **Benefit:** Reduces "Window Clipping" where a great pattern is missed because it peaked at minute 12 of a 15-minute window.

Strategy C: Linear Regression Quality (R^2)

Use statistical "Goodness of Fit" to validate the trend.

- **Concept:** Calculate a rolling linear regression line through the prices in the window.
- **Logic:** Use the **Slope** to define the magnitude and the **R-Squared (R^2)** to define the quality.
- **Benefit:** High R^2 values (e.g., > 0.85) mathematically guarantee a consistent, persistent trend rather than a single outlier spike.

Strategy D: Swing-Point (Event-Based) Detection

Abandon fixed windows in favor of market-defined structures.

- **Concept:** Identify local "Swing Highs" and "Swing Lows" (pivots).
- **Logic:** A "Bump" is the distance from a confirmed Swing Low to a Swing High. A "Slide" is the immediate reaction from that specific Swing High.
- **Benefit:** This is the most "human-like" way to trade. It adapts to the market's volatility rather than forcing the market into a 15 or 30-minute box.

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3. Technical Implementation Roadmap

Feature	Implementation Difficulty	Performance Impact
Efficiency Ratio	Low (1-2 lines of Pandas)	Negligible
Max Excursion	Low (Rolling Max)	Negligible
Linear Regression	Medium (Vectorized Matrix Math)	Moderate
Swing Points	High (Logic Restructure)	High (Iterative)

Recommendation

For the next iteration of the **Goal Seek** engine, I recommend implementing **Efficiency Ratio** and **Max Excursion** first. They provide the highest "intelligence gain" for the lowest computational cost, preserving the ultra-fast search speeds required by the project.